

Complete Tutorial: Multi-Omics Analysis of Liver Disease

A Step-by-Step Guide with Synthetic MicrobLiver Data

Author: Tutorial for Students

Date: November 2025

Purpose: Learn to analyze multi-omics longitudinal data from liver disease studies

Table of Contents

- 1. [Introduction](#)
 - 2. [Study Design](#)
 - 3. [Getting Started](#)
 - 4. [Data Structure](#)
 - 5. [Step-by-Step Analysis](#)
 - 6. [Figure Interpretation](#)
 - 7. [Code Examples](#)
 - 8. [Troubleshooting](#)
 - 9. [References](#)
-

1. Introduction

What is This Tutorial?

This tutorial teaches you how to analyze **multi-omics data** from a liver disease study. You'll learn to:

- Load and explore omics datasets (metabolites, lipids, proteins)
- Run statistical analyses (linear mixed models)
- Create publication-quality figures
- Interpret biological results

The Study

Based on the **MicrobLiver alcohol challenge study** (Israelsen et al., JHEP Reports 2021):

- **39 participants:** 14 ALD, 15 NAFLD, 10 Healthy controls
- **Alcohol challenge:** Administered via nasogastric tube over 30 minutes
- **3 time points:** 0 (baseline), 60, and 180 minutes after alcohol
- **2 sampling sites:** Hepatic (liver blood) and Systemic (peripheral blood)
- **Multi-omics:** Metabolites, lipids, proteins measured

Learning Objectives

By the end of this tutorial, you will be able to:

- 1. ✓ Load and validate omics datasets
- 2. ✓ Run linear mixed models for longitudinal data
- 3. ✓ Create 7 types of figures
- 4. ✓ Interpret multi-omics results
- 5. ✓ Understand network biology

2. Study Design

Study Overview



39 Participants

- 14 ALD (Alcohol-related Liver Disease)
- 15 NAFLD (Metabolic dysfunction-Associated Fatty Liver Disease)
- 10 Healthy controls

Alcohol Challenge (nasogastric tube, 30 min)



Blood sampling at 3 time points:

- 0 min (Baseline)
- 60 min
- 180 min

From 2 sites:

- Hepatic (liver vein)
- Systemic (peripheral vein)

Total: 39 patients × 3 times × 2 sites = 234 observations

Why This Design?

Longitudinal design: Track changes over time in same individuals

- Controls for inter-individual variation
- Reveals temporal dynamics
- Requires linear mixed models (not simple t-tests)

Dual sampling sites: Liver vs peripheral blood

- Hepatic = liver-specific metabolism
- Systemic = whole-body effects
- Can identify liver-specific changes

Multi-omics integration: Multiple molecular layers

- Metabolites (34 features) = small molecules, biochemical reactions
- Lipids (283 features) = fats, membrane components, signaling
- Proteins (77+150 features) = enzymes, regulators, structure

3. Getting Started

Required Files

Download these files from the package:

Data files (in preprocessData/ folder):

- plasma_metabolites_matrix.rds (63 KB)
- plasma_lipidomics_matrix.rds (501 KB)
- plasma_proteomics_matrix.rds (139 KB)
- plasma_MS_proteomics_matrix.rds (267 KB)

Or use Excel versions (in excel_data/ folder):

- plasma_metabolites_matrix.xlsx
- plasma_lipidomics_matrix.xlsx
- plasma_proteomics_matrix.xlsx
- plasma_MS_proteomics_matrix.xlsx

Code files:

- functions.R - Analysis functions
- network_functions.R - Network visualization
- show_data_structure.R - Validation script

Required Software

R and RStudio:

- Download R: <https://cran.r-project.org/>
- Download RStudio: <https://posit.co/download/rstudio-desktop/>

R Packages:



r

```
# Install required packages
install.packages(c(
  "lmerTest",    # Linear mixed models
  "caret",      # Data preprocessing
  "ggplot2",    # Visualization
  "ggrepel",    # Label positioning
  "Hmisc",      # Correlations
  "igraph",     # Network analysis
  "ggraph",     # Network visualization
  "RColorBrewer", # Color palettes
  "circlize",   # Circular plots
  "ComplexHeatmap" # Advanced heatmaps
))
```

Python (optional, for additional figures):

- Download Python: <https://www.python.org/downloads/>
 - Install packages: `pip install pandas numpy matplotlib seaborn scipy scikit-learn networkx`
-

4. Data Structure

Understanding the Data Format

Each dataset is a **data frame** with this structure:



- Columns 1-7: Metadata
- 1. Specimenno : Unique sample ID (e.g., "ALD_001_hepatic_T0")
 - 2. SampleID : Sample identifier (e.g., "S1_hepatic_0")
 - 3. PatientID : Patient ID (e.g., "ALD_001", "NAFLD_010", "Healthy_005")
 - 4. pheno : Phenotype - factor with 3 levels: "Healthy", "ALD", "NAFLD"
 - 5. time : Time point - factor with 3 levels: "0", "60", "180"
 - 6. sampleSite : Sampling site - factor with 2 levels: "hepatic", "systemic"
 - 7. split_col : Separator column (all NA) - marks end of metadata

- Columns 8+: Molecular Features
- Metabolite_001, Metabolite_002, ... Metabolite_034
 - Lipid_001, Lipid_002, ... Lipid_283
 - Protein_001, Protein_002, ... Protein_077
 - MSProtein_001, MSProtein_002, ... MSProtein_150

Data Properties

Already preprocessed:

- ✓ Log10 transformed
- ✓ Scaled (mean=0, sd=1)
- ✓ Missing values handled
- ✓ Low-variance features removed

Factor levels are important:

- Phenotype baseline: "Healthy" (reference group)
- Time baseline: "0" (reference time)
- Site baseline: "hepatic" (reference site)

Example: First Few Rows



r

```
# Load data
data <- readRDS("preprocessData/plasma_metabolites_matrix.rds")

# View structure
head(data[, 1:10])

#      Specimenno      SampleID PatientID pheno time sampleSite split_col Metabolite_001
# 1 ALD_001_hepatic_T0 S1_hepatic_0  ALD_001  ALD   0   hepatic      NA    -0.478714
# 2 ALD_001_hepatic_T60 S1_hepatic_60 ALD_001  ALD  60   hepatic      NA     0.152235
# 3 ALD_001_hepatic_T180 S1_hepatic_180 ALD_001  ALD 180   hepatic      NA     1.933380
```

5. Step-by-Step Analysis

Step 1: Validate the Data

Purpose: Make sure data is correctly formatted



r

```
# Run validation script
source("show_data_structure.R")
```

Expected output:



PLASMA_METABOLITES

Dimensions: 234 rows x 41 columns

Sample breakdown:

- Unique patients: 39
- Measurements per patient: 6

Phenotype distribution:

Healthy	ALD	NAFLD
60	84	90

✓ PASS: Has PatientID column

✓ PASS: Has time column

✓ PASS: Has pheno column

... (all checks should pass)

Step 2: Explore the Data

Check distributions:



```
library(ggplot2)
```

```
# Load data
```

```
data <- readRDS("preprocessData/plasma_metabolites_matrix.rds")
```

```
# Plot phenotype distribution
```

```
ggplot(data, aes(x = pheno, fill = pheno)) +  
  geom_bar() +  
  labs(title = "Sample Distribution by Phenotype",  
        x = "Phenotype", y = "Count") +  
  theme_bw()
```

```
# Plot time distribution
```

```
ggplot(data, aes(x = time, fill = pheno)) +  
  geom_bar(position = "dodge") +  
  labs(title = "Samples by Time and Phenotype",  
        x = "Time (minutes)", y = "Count") +  
  theme_bw()
```

Check a few metabolites:



r

```
# Plot one metabolite over time

ggplot(data, aes(x = time, y = Metabolite_001, color = pheno, group = pheno)) +
  stat_summary(fun = mean, geom = "line", size = 1) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  labs(title = "Metabolite_001 Over Time",
       x = "Time (minutes)", y = "Concentration (scaled)") +
  theme_bw()
```

Step 3: Run Statistical Analysis

Linear Mixed Model Analysis:



r

Load functions

```
source("functions.R")
```

Set up directories

```
data_dir <- "preprocessData/"
```

```
results_dir <- "results/"
```

```
models_dir <- "models/"
```

Create directories

```
dir.create(results_dir, showWarnings = FALSE)
```

```
dir.create(models_dir, showWarnings = FALSE)
```

Run analysis on hepatic samples

```
results_hepatic <- LMM_time(  
  name = "plasma_metabolites",  
  ID = "PatientID",  
  fixedEffect = "time",  
  groupBy = "pheno",  
  data_dir = data_dir,  
  results_dir = results_dir,  
  models_dir = models_dir,  
  sampleSite = "hepatic",  
  overwrite = TRUE,  
  padj_method = "fdr"  
)
```

Check results

```
head(results_hepatic)
```

What this does:

1. Loads metabolite data for hepatic samples
2. For each phenotype (Healthy, ALD, NAFLD):
 - For each metabolite:
 - Fits a linear mixed model: $\text{metabolite} \sim \text{time} + (1|\text{PatientID})$
 - Tests if metabolite changes at 60 min vs baseline
 - Tests if metabolite changes at 180 min vs baseline
3. Corrects p-values for multiple testing (FDR)
4. Saves results to results/results_plasma_metabolites_hepatic.rds

Interpret results:



r

How many significant features?

```
sig_results <- results_hepatic[results_hepatic$padj < 0.05, ]  
nrow(sig_results)
```

Top 10 most significant

```
top10 <- results_hepatic[order(results_hepatic$padj), ]  
head(top10[, c("molecule", "fixed_effect", "Estimate", "pvalue", "padj")], 10)
```

Step 4: Create Figures

4.1 Volcano Plot:



r

Create volcano plots

```
plots_volcano <- volcanoPlot_time(results_hepatic, pvalue = "padj")
```

View one plot

```
print(plots_volcano[["Healthy_60vs0"]])
```

Save all plots

```
for(plot_name in names(plots_volcano)) {  
  ggsave(filename = paste0("figures/volcano_", plot_name, ".png"),  
    plot = plots_volcano[[plot_name]],  
    width = 8, height = 6, dpi = 300)  
}
```

4.2 Network Visualization:



r

```
# Load network functions
```

```
source("network_functions.R")
```

```
# Load all omics data
```

```
matrix_list <- list(  
  plasma_metabolites = readRDS(paste0(data_dir, "plasma_metabolites_matrix.rds")),  
  plasma_lipidomics = readRDS(paste0(data_dir, "plasma_lipidomics_matrix.rds")),  
  plasma_proteomics = readRDS(paste0(data_dir, "plasma_proteomics_matrix.rds"))  
)
```

```
# Create network for ALD at baseline
```

```
network_ald <- getMultiOmicsNetwork(  
  matrix_list = matrix_list,  
  pheno = "ALD",  
  time = "0",  
  cor_cutoff = 0.7,  
  pval_cutoff = 0.05,  
  figures_path = "figures/",  
  save = TRUE  
)
```

```
# View the network
```

```
print(network_ald$plot)
```

```
# Check network statistics
```

```
print(network_ald$stats)
```

6. Figure Interpretation

Figure 1: PCA Plot

Show Image

What it shows: Principal Component Analysis showing how samples cluster

How to read it:

- Each point = one sample (patient at one time point)
- Colors = phenotypes (Green=Healthy, Red=ALD, Orange=NAFLD)
- PC1 (x-axis) = first principal component (explains most variance)
- PC2 (y-axis) = second principal component

What to look for:

- ✓ **Group separation:** Do phenotypes cluster separately?
- ✓ **Time progression:** Do points move along an axis over time?
- ✓ **Outliers:** Any samples far from their group?

Interpretation:

- If groups separate well → metabolic profiles are different
 - If groups overlap → similar metabolic states
 - Time progression → metabolic changes over 180 minutes
-

Figure 2: Volcano Plot

Show Image

What it shows: Statistical significance vs. effect size for each metabolite

How to read it:

- X-axis = Effect size (fold change between groups)
- Y-axis = $-\log_{10}(\text{p-value})$ (higher = more significant)
- Each point = one metabolite
- Colors: Red = upregulated, Blue = downregulated, Gray = not significant
- Horizontal line = p-value threshold ($p=0.05$)

What to look for:

- ✓ Points above horizontal line = significant
- ✓ Points far from vertical line ($x=0$) = large effect
- ✓ Labeled points = most significant features

Interpretation:

- Top-right = significantly increased metabolites
 - Top-left = significantly decreased metabolites
 - Many significant points = major metabolic changes
-

Figure 3: Heatmap

Show Image

What it shows: Concentration patterns of significant metabolites across conditions

How to read it:

- Rows = metabolites (top 20 most significant)
- Columns = conditions (phenotype × time combinations)
- Colors: Red = high concentration, Blue = low concentration
- Dendrograms show clustering

What to look for:

- ✓ Row clusters = co-regulated metabolites
- ✓ Column clusters = similar metabolic states
- ✓ Color patterns = metabolite trends

Interpretation:

- Metabolites that cluster together may be in the same pathway
 - Conditions that cluster together have similar metabolism
 - Red → blue pattern = metabolite decreases over time
-

Figure 4: Time Course Plot

Show Image

What it shows: How individual metabolites change over 180 minutes

How to read it:

- X-axis = Time (0, 60, 180 minutes)
- Y-axis = Metabolite concentration
- Lines = mean trajectory for each phenotype
- Shaded areas = standard error (uncertainty)

What to look for:

- ✓ Line slopes = rate of change
- ✓ Diverging lines = phenotype-specific responses
- ✓ Parallel lines = similar temporal dynamics

Interpretation:

- Rising line = metabolite increases after alcohol
 - Declining line = metabolite decreases after alcohol
 - Flat line = no change
 - Different slopes per phenotype = disease-specific responses
-

Figure 5: Boxplot

Show Image

What it shows: Distribution of metabolite levels across groups

How to read it:

- Boxes show quartiles (25th, 50th, 75th percentiles)
- Whiskers extend to min/max (or 1.5×IQR)
- Dots = individual data points
- Colors = phenotypes

What to look for:

- ✓ Box position = median level
- ✓ Box height = variability
- ✓ Overlapping boxes = similar distributions
- ✓ Non-overlapping = different distributions

Interpretation:

- Higher boxes = higher metabolite levels
 - Wider boxes = more variable
 - Separated boxes = significant difference
-

Figure 6: Network Graph

Show Image

What it shows: Correlation networks showing which metabolites are connected

How to read it:

- **Nodes (circles)** = metabolites
 - Size = number of connections (degree)
 - Color = phenotype group
- **Edges (lines)** = significant correlations
 - Red = positive correlation
 - Blue = negative correlation
 - Thickness = correlation strength

What to look for:

- ✓ **Network density:** Many edges = coordinated metabolism
- ✓ **Hub nodes:** Large nodes = key regulatory molecules
- ✓ **Modules:** Clusters of connected nodes = functional groups
- ✓ **Differences:** Compare across phenotypes

Interpretation (from this figure):

- **Healthy:** 7 nodes, 4 edges - simple, coordinated
- **ALD:** 2 nodes, 1 edge - **disrupted metabolism** (fewest connections!)
- **NAFLD:** 13 nodes, 9 edges - most complex network

Key finding: ALD shows the most metabolic disruption (simplest network), while NAFLD maintains more metabolic complexity. This suggests different disease mechanisms!

Figure 7: Bar Chart

Show Image

What it shows: Summary of how many metabolites are up/down-regulated

How to read it:

- X-axis = Comparison groups
- Y-axis = Number of significant features
- Red bars = upregulated (increased)
- Blue bars = downregulated (decreased)
- Numbers on bars = exact counts

What to look for:

- ✓ Bar heights = number of changes
- ✓ Red vs blue balance = direction of changes
- ✓ Compare across groups

Interpretation:

- Taller bars = more metabolic changes
 - More red = predominantly increases
 - More blue = predominantly decreases
 - Equal red/blue = balanced response
-

7. Code Examples

Complete Analysis Workflow



r

```

# =====
# COMPLETE ANALYSIS SCRIPT
# Multi-Omics Analysis of Liver Disease Study
# =====

# 1. SETUP
# =====

library(lmerTest)
library(caret)
library(ggplot2)
library(ggrepel)

source("functions.R")
source("network_functions.R")

data_dir <- "preprocessData/"
results_dir <- "results/"
models_dir <- "models/"
figures_dir <- "figures/"

dir.create(results_dir, showWarnings = FALSE, recursive = TRUE)
dir.create(models_dir, showWarnings = FALSE, recursive = TRUE)
dir.create(figures_dir, showWarnings = FALSE, recursive = TRUE)

# 2. LOAD AND VALIDATE DATA
# =====

cat("\n=== LOADING DATA ===\n")

# Validate data structure
source("show_data_structure.R")

# Load metabolites
metabolites <- readRDS(paste0(data_dir, "plasma_metabolites_matrix.rds"))
cat("Loaded metabolites:", nrow(metabolites), "samples\n")

# 3. RUN STATISTICAL ANALYSIS
# =====

cat("\n=== RUNNING LINEAR MIXED MODELS ===\n")

# Analyze hepatic samples
results_hepatic <- LMM_time(
  name = "plasma_metabolites",
  ID = "PatientID",
  fixedEffect = "time",

```

```

groupBy = "pheno",
data_dir = data_dir,
results_dir = results_dir,
models_dir = models_dir,
sampleSite = "hepatic",
overwrite = TRUE
)

cat("Results:", nrow(results_hepatic), "tests\n")
cat("Significant (padj<0.05):", sum(results_hepatic$padj < 0.05, na.rm=TRUE), "\n")

# 4. CREATE VOLCANO PLOTS
# =====
cat("\n=== CREATING VOLCANO PLOTS ===\n")

plots_volcano <- volcanoPlot_time(results_hepatic, pvalue = "padj")

for(plot_name in names(plots_volcano)) {
  ggsave(
    filename = paste0(figures_dir, "volcano_", plot_name, ".png"),
    plot = plots_volcano[[plot_name]],
    width = 8, height = 6, dpi = 300
  )
  cat("Saved:", plot_name, "\n")
}

# 5. CREATE NETWORK GRAPHS
# =====
cat("\n=== CREATING NETWORK VISUALIZATIONS ===\n")

# Load all omics layers
matrix_list <- list(
  plasma_metabolites = readRDS(paste0(data_dir, "plasma_metabolites_matrix.rds")),
  plasma_lipidomics = readRDS(paste0(data_dir, "plasma_lipidomics_matrix.rds")),
  plasma_proteomics = readRDS(paste0(data_dir, "plasma_proteomics_matrix.rds"))
)

# Create networks for each phenotype at baseline
for(pheno in c("Healthy", "ALD", "NAFLD")) {
  cat("\nCreating network for", pheno, "...")

  network <- getMultiOmicsNetwork(
    matrix_list = matrix_list,
    pheno = pheno,

```



```

time = "0",
cor_cutoff = 0.7,
pval_cutoff = 0.05,
figures_path = figures_dir,
save = TRUE
)

if(!is.null(network)) {
  cat(" Nodes:", network$stats$n_nodes, "\n")
  cat(" Edges:", network$stats$n_edges, "\n")
}
}

# 6. SUMMARY STATISTICS
# =====

cat("\n=== SUMMARY ===\n")

# Count significant features per phenotype
summary_table <- data.frame()

for(pheno in c("Healthy", "ALD", "NAFLD")) {
  for(time in c("60vs0", "180vs0")) {
    pattern <- paste0(pheno, "_", time)

    subset <- results_hepatic[grepl(pattern, results_hepatic$fixed_effect), ]
    n_sig <- sum(subset$padj < 0.05, na.rm = TRUE)
    n_up <- sum(subset$padj < 0.05 & subset$Estimate > 0, na.rm = TRUE)
    n_down <- sum(subset$padj < 0.05 & subset$Estimate < 0, na.rm = TRUE)

    summary_table <- rbind(summary_table, data.frame(
      Phenotype = pheno,
      Comparison = time,
      Total_Significant = n_sig,
      Upregulated = n_up,
      Downregulated = n_down
    ))
  }
}

print(summary_table)

# Save summary
write.csv(summary_table,
  paste0(results_dir, "summary_statistics.csv"),

```

```
row.names = FALSE)
```

```
cat("\n=== ANALYSIS COMPLETE ===\n")
```

```
cat("Results saved to:", results_dir, "\n")
```

```
cat("Figures saved to:", figures_dir, "\n")
```

Quick Analysis Snippets

Load and explore one dataset:



r

```
# Load data
```

```
data <- readRDS("preprocessData/plasma_metabolites_matrix.rds")
```

```
# Check dimensions
```

```
dim(data) # Should be 234 x 41
```

```
# Check structure
```

```
str(data)
```

```
# Summary statistics
```

```
summary(data[, c("pheno", "time", "sampleSite")])
```

```
# View first few rows
```

```
head(data[, 1:10])
```

Extract significant metabolites:



r

Load results

```
results <- readRDS("results/results_plasma_metabolites_hepatic.rds")
```

Get significant features

```
sig_features <- results[results$padj < 0.05 & !grepl("Intr", results$fixed_effect), ]
```

Order by p-value

```
sig_features <- sig_features[order(sig_features$padj), ]
```

Top 20

```
top20 <- head(sig_features, 20)
```

```
print(top20[, c("molecule", "fixed_effect", "Estimate", "pvalue", "padj")])
```

Create custom plot:



r

Load data

```
data <- readRDS("preprocessData/plasma_metabolites_matrix.rds")
```

Get one significant metabolite

```
metabolite <- "Metabolite_005"
```

Plot trajectory

```
ggplot(data, aes(x = time, y = .data[[metabolite]],
                 color = pheno, group = pheno)) +
  stat_summary(fun = mean, geom = "line", size = 1) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2) +
  scale_color_manual(values = c("Healthy" = "#2ca02c",
                                "ALD" = "#d62728",
                                "NAFLD" = "#ff7f0e")) +
  labs(title = paste("Trajectory of", metabolite),
       x = "Time (minutes)",
       y = "Concentration (scaled)",
       color = "Phenotype") +
  theme_bw() +
  theme(legend.position = "bottom")

ggsave("figures/custom_trajectory.png", width = 8, height = 6, dpi = 300)
```

8. Troubleshooting

Common Issues and Solutions

Issue 1: "Cannot find file"

Error: Error in readRDS("plasma_metabolites_matrix.rds") : cannot open the connection

Solution:



r

```
# Check your working directory
getwd()

# Set working directory to where files are located
setwd("/path/to/your/files")

# Or use full paths
data <- readRDS("/full/path/to/preprocessData/plasma_metabolites_matrix.rds")
```

Issue 2: "Package not found"

Error: Error in library(lmerTest) : there is no package called 'lmerTest'

Solution:



r

```
# Install missing package
install.packages("lmerTest")

# Load it
library(lmerTest)
```

Issue 3: "Factor levels incorrect"

Error: Results don't make sense or contrasts are wrong

Solution:



r

Check factor levels

```
levels(data$pheno) # Should be: "Healthy" "ALD" "NAFLD"
```

```
levels(data$time) # Should be: "0" "60" "180"
```

If wrong, reload data (don't modify manually)

```
data <- readRDS("preprocessData/plasma_metabolites_matrix.rds")
```

Issue 4: "No significant results"

Problem: All p-values > 0.05

Solutions:

- This is expected with small synthetic data
- Check sample size: `table(data$pheno, data$time)`
- Try less stringent threshold: `padj < 0.1`
- With real data, you'll have more power

Issue 5: "Network is empty"

Problem: Network plot shows no nodes or edges

Solutions:



r

Lower correlation threshold

```
network <- getMultiOmicsNetwork(  
  matrix_list,  
  pheno = "ALD",  
  time = "0",  
  cor_cutoff = 0.5, # Lower from 0.7  
  pval_cutoff = 0.1 # Less stringent  
)
```

Use more features

```
network <- getMultiOmicsNetwork(  
  matrix_list,  
  pheno = "ALD",  
  time = "0",  
  top_n_per_omics = 50 # Increase from 30  
)
```

Issue 6: "Plot labels overlap"

Problem: Can't read labels in volcano plot

Solution:



r

```
# The ggrepel package automatically handles this
# But you can adjust parameters:

# In volcanoPlot_time function, modify:
geom_label_repel(aes(label = label),
  color = "black",
  size = 2.5,      # Smaller text
  box.padding = 0.5, # More padding
  max.overlaps = 20) # Allow more labels
```

Getting Help

If you're stuck:

1. Read error messages carefully
2. Check the documentation files (README.md, NETWORK_GUIDE.md)
3. Google the error message
4. Check R documentation: ?function_name
5. Ask for help with reproducible example

9. Expected Results

What You Should Get

Running the complete analysis on the synthetic data, you should obtain:

Statistical Results:

- ~20-40 significant metabolites ($p_{adj} < 0.05$)
- Mix of up- and down-regulated features
- More changes at 180 min than 60 min
- Different profiles for ALD vs NAFLD

Figures Generated:

1. ✓ PCA plots (3 panels for 3 time points)
2. ✓ Volcano plots (6 plots: 3 phenotypes $\times$ 2 time comparisons)
3. ✓ Heatmap (1 plot with top features)
4. ✓ Time course plots (6 panels with trajectories)
5. ✓ Boxplots (4 panels for significant features)
6. ✓ Network graphs (3 plots for 3 phenotypes)
7. ✓ Bar chart (1 summary plot)

Network Results:

- Healthy: ~5-10 nodes, sparse network

- ALD: ~2-5 nodes, very sparse (metabolic disruption)
- NAFLD: ~10-15 nodes, denser network
- Positive and negative correlations
- Some hub molecules (high degree)

Files Created:



```

results/
|—— results_plasma_metabolites_hepatic.rds
|—— results_plasma_metabolites_systemic.rds
|—— summary_statistics.csv

models/
|—— models_plasma_metabolites_hepatic.rds
|—— models_plasma_metabolites_systemic.rds

figures/
|—— volcano_Healthy_60vs0.png
|—— volcano_Healthy_180vs0.png
|—— volcano_ALD_60vs0.png
|—— volcano_ALD_180vs0.png
|—— volcano_NAFLD_60vs0.png
|—— volcano_NAFLD_180vs0.png
|—— network_Healthy_0.png
|—— network_ALD_0.png
|—— network_NAFLD_0.png

```

10. Real Data Considerations

When You Use Your Real Data

Differences from synthetic data:

1. **More significant results:** Real biology has stronger signals
2. **Clearer patterns:** Better group separation in PCA
3. **Denser networks:** More biological correlations
4. **Known molecules:** Real lipid/metabolite names, not generic labels
5. **Biological interpretation:** Can map to known pathways

Additional analyses to do:

1. **Pathway enrichment:** Map features to KEGG/Reactome pathways
2. **Biological validation:** Check literature for known markers
3. **Clinical correlations:** Link to patient outcomes if available
4. **Hepatic vs systemic comparison:** Identify liver-specific changes
5. **Multi-omics integration:** Correlate across molecular layers

Publication tips:

- Use volcano plots for main results
 - Network graphs show biological relationships
 - Time courses tell the temporal story
 - Heatmaps give comprehensive overview
 - Always report: n, statistics, correction method
-

11. Key Concepts Review

Statistical Concepts

Linear Mixed Models:

- Fixed effects: Time (what we want to test)
- Random effects: Patient ID (accounts for repeated measures)
- Why not t-tests? Because measurements from same patient are correlated

Multiple Testing Correction:

- Testing many features (34 metabolites, 283 lipids, etc.)
- Without correction: 5% false positives per test
- FDR (False Discovery Rate): Controls proportion of false discoveries
- Adjusted p-value (p_{adj}) < 0.05 is significant

Effect Size vs Significance:

- P-value: Is the difference real? (statistical significance)
- Effect size: How big is the difference? (biological importance)
- Need both for interpretation!

Biological Concepts

Metabolomics: Small molecules in biological systems

- Examples: glucose, amino acids, TCA cycle intermediates
- Reflect active biochemical processes
- Fast turnover, immediate responses

Lipidomics: Lipid molecules

- Examples: phospholipids, triglycerides, sphingolipids
- Cell membranes, energy storage, signaling
- Slower turnover than metabolites

Proteomics: Proteins

- Enzymes, receptors, structural proteins
- Execute biological functions
- Slowest turnover

Multi-omics Integration: Connecting the layers

- Proteins (enzymes) → catalyze reactions
 - Metabolites (substrates/products) → consumed/produced
 - Lipids → structural and regulatory roles
 - Networks show these relationships
-

12. Learning Exercises

Exercise 1: Basic Analysis

Task: Analyze systemic samples instead of hepatic



r

```
# Run analysis on systemic samples
results_systemic <- LMM_time(
  name = "plasma_metabolites",
  ID = "PatientID",
  fixedEffect = "time",
  groupBy = "pheno",
  data_dir = data_dir,
  results_dir = results_dir,
  models_dir = models_dir,
  sampleSite = "systemic", # Changed from "hepatic"
  overwrite = TRUE
)

# Compare results
n_sig_hepatic <- sum(results_hepatic$padj < 0.05, na.rm=TRUE)
n_sig_systemic <- sum(results_systemic$padj < 0.05, na.rm=TRUE)

cat("Significant in hepatic:", n_sig_hepatic, "\n")
cat("Significant in systemic:", n_sig_systemic, "\n")
```

Question: Do you find more changes in hepatic or systemic samples? Why might that be?

Exercise 2: Time Comparison

Task: Compare 60 min vs 180 min results



r

Extract results for each time point

```
results_60 <- results_hepatic[grepl("60vs0", results_hepatic$fixed_effect), ]  
results_180 <- results_hepatic[grepl("180vs0", results_hepatic$fixed_effect), ]
```

Count significant features

```
sig_60 <- sum(results_60$padj < 0.05, na.rm=TRUE)  
sig_180 <- sum(results_180$padj < 0.05, na.rm=TRUE)
```

```
cat("Significant at 60 min:", sig_60, "\n")  
cat("Significant at 180 min:", sig_180, "\n")
```

Question: Are there more changes at 60 or 180 minutes? What does this tell you about the metabolic response?

Exercise 3: Multi-Omics Analysis

Task: Run the same analysis on lipidomics data



r

Analyze lipids

```
results_lipids <- LMM_time(  
  name = "plasma_lipidomics", # Changed dataset  
  ID = "PatientID",  
  fixedEffect = "time",  
  groupBy = "pheno",  
  data_dir = data_dir,  
  results_dir = results_dir,  
  models_dir = models_dir,  
  sampleSite = "hepatic",  
  overwrite = TRUE  
)
```

Compare with metabolites

```
cat("Significant metabolites:", sum(results_hepatic$padj < 0.05, na.rm=TRUE), "\n")  
cat("Significant lipids:", sum(results_lipids$padj < 0.05, na.rm=TRUE), "\n")
```

Question: Do you see more changes in metabolites or lipids? Why?

Exercise 4: Network Comparison

Task: Compare networks at different time points



r

```
# Create networks at baseline and 180 min
network_baseline <- getMultiOmicsNetwork(
  matrix_list, pheno = "ALD", time = "0",
  cor_cutoff = 0.6, save = FALSE
)

network_180 <- getMultiOmicsNetwork(
  matrix_list, pheno = "ALD", time = "180",
  cor_cutoff = 0.6, save = FALSE
)

# Compare
cat("Baseline - Nodes:", network_baseline$stats$n_nodes,
    "Edges:", network_baseline$stats$n_edges, "\n")
cat("180 min - Nodes:", network_180$stats$n_nodes,
    "Edges:", network_180$stats$n_edges, "\n")
```

Question: How does the network change after alcohol? What does this mean biologically?

13. References

Original Study

Israelsen M, Kim M, Suvitaival T, et al. Comprehensive lipidomics reveals phenotypic differences in hepatic lipid turnover in ALD and NAFLD during alcohol intoxication. *JHEP Reports*. 2021;3(5):100325. doi: 10.1016/j.jhepr.2021.100325

Statistical Methods

- Bates D, et al. Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software*. 2015;67(1):1-48.
- Kuznetsova A, et al. lmerTest Package: Tests in Linear Mixed Effects Models. *Journal of Statistical Software*. 2017;82(13):1-26.

Multi-Omics Integration

- Hasin Y, et al. Multi-omics approaches to disease. *Genome Biology*. 2017;18:83.
- Subramanian I, et al. Multi-omics Data Integration, Interpretation, and Its Application. *Bioinformatics and Biology Insights*. 2020;14:1177932219899051.

Network Analysis

- Barabási AL, et al. Network medicine: a network-based approach to human disease. *Nature Reviews Genetics*. 2011;12:56-68.
- Csardi G, Nepusz T. The igraph software package for complex network research. *InterJournal, Complex Systems*. 2006;1695.

R Resources

- R Core Team. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>
 - Wickham H. ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag New York, 2016.
-

14. Appendix: Glossary

ALD: Alcohol-related Liver Disease - liver damage caused by excessive alcohol consumption

NAFLD/MASLD: Non-Alcoholic Fatty Liver Disease / Metabolic dysfunction-Associated Steatotic Liver Disease - liver fat accumulation not caused by alcohol

Omics: Large-scale molecular profiling (metabolomics, lipidomics, proteomics, genomics)

Linear Mixed Model: Statistical model with both fixed effects (variables of interest) and random effects (accounting for repeated measures)

FDR: False Discovery Rate - method to control false positives in multiple testing

PCA: Principal Component Analysis - dimension reduction technique to visualize high-dimensional data

Volcano Plot: Plot showing effect size vs statistical significance

Network Graph: Visualization of relationships (correlations) between molecules

Hub Molecule: Highly connected node in a network, often a key regulatory molecule

Hepatic: Related to the liver

Systemic: Related to whole-body circulation

Longitudinal Study: Following same subjects over time

Effect Size: Magnitude of difference between groups

Pathway: Series of molecular interactions leading to a biological outcome

15. Checklist for Students

Before you finish, make sure you can:

- ☐ Load the synthetic datasets in R
- ☐ Understand the study design (39 patients, 3 times, 2 sites)
- ☐ Run the validation script successfully
- ☐ Execute the LMM_time analysis
- ☐ Create at least 3 types of figures
- ☐ Interpret a volcano plot
- ☐ Interpret a network graph
- ☐ Explain what a linear mixed model is
- ☐ Explain why multiple testing correction is needed
- ☐ Identify significant results ($padj < 0.05$)
- ☐ Compare hepatic vs systemic results
- ☐ Compare different phenotypes (Healthy, ALD, NAFLD)
- ☐ Explain the biological meaning of your results

16. Next Steps

For Further Learning

- 1. **Advanced R Programming:**
 - Tidyverse for data manipulation
 - Advanced ggplot2 customization
 - Writing your own functions
- 2. **More Statistical Methods:**
 - Multivariate analysis (PLS-DA, OPLS-DA)
 - Machine learning for classification
 - Time series analysis
- 3. **Bioinformatics:**
 - Pathway enrichment analysis (KEGG, Reactome)
 - Gene set enrichment analysis (GSEA)
 - Integration with genomics data
- 4. **Data Science Skills:**
 - Python for data analysis
 - Version control with Git
 - Reproducible research with RMarkdown

Practice Projects

- 1. **Analyze all omics layers:** Run complete analysis on metabolites, lipids, and proteins
- 2. **Create a report:** Use RMarkdown to document your analysis
- 3. **Build a dashboard:** Use Shiny to create interactive visualizations
- 4. **Pathway analysis:** Map significant features to biological pathways
- 5. **Predictive modeling:** Build a classifier to distinguish phenotypes

Congratulations! You now have the skills to analyze multi-omics longitudinal data! 🎉

Document Version: 1.0
Last Updated: November 2025
For questions or feedback, refer to the README.md and other documentation files.